

Project 01: Multi-Agent Orchestration System using LangGraph, GitHub API, and Docker

Project 02: Production RAG Pipeline with Evaluation Layer using RAGAS, hybrid retrieval, and cross-encoder reranking

Project 03: Local LLM Deployment on Edge Hardware using Ollama, llama.cpp, and GGUF quantization

Project 04: AI-Powered Code Review CLI using Python, Pydantic structured outputs, and GitHub Actions

Project 05: Multimodal AI App with Real Users using GPT-4o Vision and vision classifier routing

1. The Autonomous Medical Triage & Diagnostic Copilot

- **The Concept:** A system that assists doctors in understaffed clinics by taking in patient histories, looking at medical scans, searching through vast medical literature, and synthesizing a preliminary diagnostic report.
- **How the Tech Blends:**
 - **Multimodal AI:** The system accepts both text (patient symptoms/history) and images (X-rays, MRIs, or even photos of skin conditions).
 - **RAG Pipeline:** It actively searches a vector database containing thousands of medical guidelines, drug interactions, and previous case studies to ensure recommendations are fact-based.
 - **Multi-Agent Orchestration:**
 - **Agent 1 (*The Vision Specialist*):** Analyzes the scan and outputs text descriptions of anomalies.
 - **Agent 2 (*The Researcher*):** Takes the symptoms and Agent 1's findings, then runs the RAG queries.
 - **Agent 3 (*The Synthesizer*):** Combines everything into a structured, readable report for the human doctor.

3. Automated Legal Contract & Risk Auditor

- **The Concept:** Corporate lawyers spend weeks manually reading massive contracts, looking for hidden liabilities or non-compliant clauses. This system ingests entire contract portfolios, visually scans them, cross-references corporate law, and redlines risky sections.
- **How the Tech Blends:**
 - **Multimodal AI:** Processes scanned PDFs, recognizing wet signatures, stamped tables, and handwritten margin notes, not just digital text.
 - **RAG Pipeline:** Grounded on a massive database of state laws, past company contracts, and compliance regulations.
 - **Multi-Agent Orchestration:**
 - **Agent 1 (The Extractor):** Pulls out key entities, dates, and obligations.
 - **Agent 2 (The Auditor):** Uses RAG to check if any extracted clause violates known compliance rules.
 - **Agent 3 (The Negotiator):** Automatically drafts an alternative, safer version of the risky clause.

4. Smart City Incident Command System

- **The Concept:** When a major city incident occurs (like a fire or traffic pileup), emergency dispatchers are overwhelmed with fragmented data. This system acts as a real-time command center, fusing different data streams to deploy resources effectively.
- **How the Tech Blends:**
 - **Multimodal AI:** Processes live CCTV video feeds (vision) and transcribes emergency audio calls (audio) simultaneously.
 - **RAG Pipeline:** Instantly retrieves city infrastructure blueprints, fire hydrant locations, and Standard Operating Procedures (SOPs) for specific disaster types.
 - **Multi-Agent Orchestration:**
 - **Agent 1 (The Watcher):** Flags anomalies in video feeds (e.g., smoke detection).
 - **Agent 2 (The Contextualizer):** Queries the RAG system for the blueprint of the affected area.

- **Agent 3 (*The Dispatcher*):** Calculates and suggests the optimal deployment of fire trucks and ambulances.

1. Urban PropTech: Automated Blueprint & Zoning Auditor

- **The Unique Real-World Problem:** Construction projects face massive delays and fines because physical blueprints often violate obscure local zoning bylaws or environmental regulations. Currently, humans have to manually cross-reference drawings against hundreds of pages of municipal codes.
- **How the Tech Blends:**
 - **Multimodal AI:** Ingests architectural CAD drawings and floor plan images, parsing the spatial dimensions and structural notes.
 - **RAG Pipeline:** Grounded in a highly specific database of municipal building codes (e.g., Gujarat's General Development Control Regulations).
 - **Multi-Agent Orchestration:**
 - **Agent 1 (*The Vision Architect*):** Extracts measurements, room types, and material notes from the blueprint image.
 - **Agent 2 (*The Compliance Officer*):** Takes Agent 1's data and queries the RAG database to find any legal violations (e.g., "fire exit is too narrow" or "building exceeds height limit").
 - **Agent 3 (*The Draftsman*):** Generates a final JSON report redlining the exact coordinates of the violations on the blueprint.

3. Agentic Agritech: The Smallholder's Market Negotiator

- **The Unique Real-World Problem:** Small-scale farmers often get exploited by middlemen because they cannot accurately grade their own crop quality or track real-time market fluctuations across different regional *mandis* (markets).
- **How the Tech Blends:**
 - **Multimodal AI:** The farmer uploads photos of a fresh crop batch to assess color, size, and defect rate (grading), alongside a voice note asking for selling advice.
 - **RAG Pipeline:** Connected to live market price sheets (like AGMARKNET), historical regional demand, and logistics cost databases.
 - **Multi-Agent Orchestration:**

- **Agent 1 (The Grader):** Analyzes the crop image and assigns an objective quality grade (e.g., Grade A vs. Grade C).
- **Agent 2 (The Market Analyst):** Uses the grade to query the RAG database for current fair-market values and transport costs to nearby cities.
- **Agent 3 (The Negotiator):** Synthesizes the data and formulates a specific target price and negotiation strategy for the farmer.

4. E-Commerce "Dark Pattern" & GenAI Scam Watchdog

- **The Unique Real-World Problem:** Dropshippers and scam sellers are using AI-generated product images, deepfake video testimonials, and LLM-written reviews to manipulate buyers. Standard review filters can't catch these complex multimodal scams.
- **How the Tech Blends:**
 - **Multimodal AI:** The user inputs a product URL. The system scrapes the product images, promotional videos, and text reviews.
 - **RAG Pipeline:** References a database of known factory-sourcing templates (like Alibaba white-label goods), baseline fake-review patterns, and consumer protection alerts.
 - **Multi-Agent Orchestration:**
 - **Agent 1 (The Visual Forensics Expert):** Scans the product images and videos specifically looking for generative AI artifacts (e.g., weird lighting, six fingers, inconsistent branding).
 - **Agent 2 (The Linguistic Auditor):** Analyzes the text reviews for bot-like repetition and sentiment manipulation.
 - **Agent 3 (The Risk Profiler):** Combines the visual anomalies and text data to assign a final "Deception Risk Score."

1. Autonomous Public Welfare & Civic Application Copilot

- **The Concept:** Upgrading a standard welfare recommendation platform into an autonomous, agentic system. Instead of a user just searching for schemes, the

system actively processes their physical documents, maps their eligibility, and drafts the applications.

- **How the Tech Blends:**
 - **Multimodal AI:** Citizens upload smartphone photos of their handwritten utility bills, physical IDs, or proof of crop damage. The vision model extracts the structured data.
 - **GraphRAG Pipeline:** A Knowledge Graph maps the complex relationships between regional welfare policies, income thresholds, and required documentation.
 - **Multi-Agent Orchestration:**
 - **Agent 1 (The Extractor):** Parses the uploaded images and structures the citizen's profile.
 - **Agent 2 (The Eligibility Auditor):** Navigates the GraphRAG network to find exact scheme matches without hallucinating.
 - **Agent 3 (The Filer):** Autonomously drafts the localized application form, ready for submission.
- **The Execution:** You can build the entire interactive frontend using Streamlit, giving it a clean, accessible UI, while orchestrating the agents with LangGraph in the backend.

1. Autonomous Intellectual Property & Brand Defender

- **The Real-World Problem:** E-commerce platforms and social media are flooded with counterfeit products and stolen designs. Brands cannot manually hire enough lawyers to watch millions of video reviews or scan marketplace listings to issue takedowns.
- **How the Tech Blends:**
 - **Multimodal AI:** Scrapes and processes marketplace images, promotional videos, and audio tracks to detect visual brand impersonation or stolen product designs.
 - **RAG Pipeline:** Grounded in a highly specific vector database of the brand's registered design patents, official CAD schematics, and regional copyright laws.
 - **Multi-Agent Orchestration:**

- ***Agent 1 (The Visual Forensics Expert):*** Analyzes the scraped media to identify unauthorized logos or stolen structural designs.
- ***Agent 2 (The Legal Auditor):*** Cross-references Agent 1's findings against the RAG database to confirm if a specific design patent is being violated.
- ***Agent 3 (The Enforcer):*** Automatically drafts a legally sound Cease & Desist notice or a structured DMCA takedown payload for the hosting platform.